testbench.sv

```
module tb;
    reg  [3:0] a, b;
    reg  [1:0] select;
    wire [3:0] cikis;

    alu uut (.a(a), .b(b), .select(select), .cikis(cikis));

    initial begin
        $dumpfile("dump.vcd");
        $dumpvars(0, tb);

        a = 4'b0110; b = 4'b0011;   // a=6, b=3

        select = 2'b00; #10;
        $display("select=%b a=%b b=%b cikis=%b (toplama)", select, a, b, cikis);

        select = 2'b01; #10;
        $display("select=%b a=%b b=%b cikis=%b (cikarma)", select, a, b, cikis);

        select = 2'b10; #10;
        $display("select=%b a=%b b=%b cikis=%b (AND)", select, a, b, cikis);

        select = 2'b11; #10;
        $display("select=%b a=%b b=%b cikis=%b (OR)", select, a, b, cikis);

        $finish;
    end
endmodule
```

design.sv

```
module alu (
    input  [3:0] a, b,
    input  [1:0] select,
    output reg [3:0] cikis
);
    always @(*) begin
        case (select)
            2'b00: cikis = a + b;   // toplama
            2'b01: cikis = a - b;   // cikarma
            2'b10: cikis = a & b;   // AND
            2'b11: cikis = a | b;   // OR
        endcase
    end
endmodule
```

```
select=00 a=0110 b=0011 cikis=1001 (toplama)
select=01 a=0110 b=0011 cikis=0011 (cikarma)
select=10 a=0110 b=0011 cikis=0010 (AND)
select=11 a=0110 b=0011 cikis=0111 (OR)
testbench.sv:26: $finish called at 40 (1s)
```

